

A three-generator monomial ideal whose v -number increases under integral closure

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Abstract

For every field K , the monomial ideal $J = (y^2, yz^3, x^3z^4) \subset K[x, y, z]$ has exactly three minimal monomial generators and satisfies $v(J) = 3 < 4 = v(\overline{J})$. Its integral closure is $(y^2, yz^3, x^2yz^2, x^3z^4)$. We give a direct proof of the closure formula and of both v -numbers, allowing arbitrary homogeneous polynomial witnesses and all associated primes. Thus a bound of three on the number of generators does not ensure that integral closure decreases the v -number.

1 The comparison and the example

Let $R = K[x, y, z]$ have the standard grading. For a proper homogeneous ideal H , write $(H : f) = \{g \in R : gf \in H\}$ and define

$$v(H) = \min\{\deg f : 0 \neq f \in R \text{ homogeneous, } (H : f) = \mathfrak{p} \in \text{Ass}(R/H)\}.$$

Here $\text{Ass}(R/H)$ includes embedded associated primes. A proper prime colon $(H : f)$ is the annihilator of the nonzero class of f in R/H , so it is an associated prime by definition. Write $\overline{H}$ for the integral closure of H .

Biswas, Mandal, and Phukan prove the comparison $v(\overline{H}) \leq v(H)$ for monomial ideals in two variables [1, Theorem 3.2]. They also prove its analogue for every positive power of an equigenerated monomial ideal in three variables [1, Theorem 3.12]. Their Example 3.3 exhibits the failure

$$H = (x^2, y^2, z^5, xyz), \quad v(H) = 2 < 3 = v(\overline{H}).$$

Vanmathi A and Sarkar establish the comparison for all positive powers of monomial complete intersections [2, Theorem 4.14(1)(i)]. We consider a generator-count refinement of the comparison problem: does $v(\overline{H}) \leq v(H)$ hold for every proper monomial ideal with at most three minimal generators, over every field and in any number of variables? The following example answers this question negatively.

Theorem 1. *For every field K , the ideal*

$$J = (y^2, yz^3, x^3z^4) \subset R = K[x, y, z]$$

is proper and has exactly three minimal monomial generators. Moreover,

$$\overline{J} = Q := (y^2, yz^3, x^2yz^2, x^3z^4), \quad v(J) = 3, \quad v(Q) = 4.$$

The three displayed generators of J have degrees 2, 4, 7 and none divides another. Thus the example is not equigenerated. The contribution relative to the four-generator example above is failure with three generators. We make no assertion about two generators, minimal exponent sizes, higher powers of J , or publication priority.

2 Integral closure

We use the standard monomial integral-closure criterion, recalled in [1, Introduction]: if a monomial ideal has generator exponent vectors $A_1, \dots, A_r$, then its integral closure is monomial and its exponents are the lattice points of

$$\text{NP}(H) = \text{conv}\{A_1, \dots, A_r\} + \mathbb{R}_{\geq 0}^3.$$

For the present ideal the vectors are $A = (0, 2, 0)$, $B = (0, 1, 3)$, and $C = (3, 0, 4)$. Put $t = x^2yz^2$. Since

$$t^2 = x(y^2)(x^3z^4) \in J^2,$$

the equation $T^2 - t^2 = 0$ shows that t is integral over J . Consequently $Q \subseteq \bar{J}$.

For the reverse inclusion, every point (a, b, c) of $\text{NP}(J)$ satisfies

$$a + 3b \geq 3, \quad 2b + c \geq 4, \quad 2a + 9b + 3c \geq 18. \quad (1)$$

Indeed the three linear forms have nonnegative coefficients; their values at A, B, C are respectively $(6, 3, 3)$, $(4, 5, 4)$, and $(18, 18, 18)$. Their lower bounds are therefore preserved by convex combinations and addition of nonnegative vectors.

Now let (a, b, c) be a nonnegative integer point satisfying (1). If $b \geq 2$, its monomial is divisible by y^2 . If $b = 0$, the first two inequalities give $a \geq 3$ and $c \geq 4$, so it is divisible by x^3z^4 . If $b = 1$, the second inequality gives $c \geq 2$. When $c \geq 3$, the monomial is divisible by yz^3 ; when $c = 2$, the third inequality gives $2a \geq 3$, hence $a \geq 2$, and the monomial is divisible by x^2yz^2 . These cases prove $\bar{J} \subseteq Q$. Only necessary inequalities for the real Newton polyhedron were used; the integer case distinction applies to all nonnegative exponents.

3 The two v -numbers

For monomial generators u_i and a monomial w , divisibility gives

$$((u_1, \dots, u_r) : w) = (u_i / \gcd(u_i, w) : 1 \leq i \leq r). \quad (2)$$

In particular,

$$(J : yz^2) = (y, z), \quad (Q : xyz^2) = (x, y, z). \quad (3)$$

The witnesses lie outside their respective ideals, and both colons are proper prime ideals over every field. Hence $v(J) \leq 3$ and $v(Q) \leq 4$.

For the lower bounds we work directly with homogeneous polynomials. If $(H : f)$ is proper and H is monomial, deleting from f all terms belonging to H preserves the colon and leaves a nonzero homogeneous polynomial of the same degree. A polynomial belongs to a monomial ideal exactly when every term does. Multiplication by a fixed monomial sends distinct exponent vectors to distinct exponent vectors, so no cancellation occurs in any characteristic.

The lower bound for J . Suppose $\deg f \leq 2$ and $\mathfrak{p} = (J : f)$ is prime, after the deletion just described. Since $y^2 \in J \subseteq \mathfrak{p}$, primality gives $y \in \mathfrak{p}$, so $yf \in J$. A surviving term $x^a y^b z^c$ of f must have $b = 1$: terms with $b \geq 2$ were deleted, while for $b = 0$ the membership $yx^a z^c \in J$ would require at least $c \geq 3$, exceeding the degree bound. Thus $f = yg(x, z)$ with $g \neq 0$ and $\deg g \leq 1$. Now $z^3 f \in J$, so primality forces $zf \in J$. But every term of zf has y -exponent one and z -exponent at most two, and is divisible by none of the generators of J . This contradiction proves $v(J) \geq 3$.

The lower bound for Q . Suppose instead that $\deg f \leq 3$ and $\mathfrak{p} = (Q : f)$ is prime. Again $yf \in Q$. After deleting ideal terms, the possible terms of f are

$$x^a y z^c \quad (a + c \leq 2), \quad \text{and possibly } z^3. \quad (4)$$

Indeed a term with y -exponent zero can enter Q after multiplication by y only if $c \geq 3$, or $a \geq 2, c \geq 2$, or $a \geq 3, c \geq 4$. Only z^3 fits the degree bound.

If z^3 occurs with nonzero coefficient, neither xf nor zf belongs to Q , because the terms xz^3 and z^4 , respectively, survive. On the other hand $x^3 z^2 f \in Q$: its product with every term of the first type in (4) is divisible by $x^2 y z^2$, and its product with z^3 is divisible by $x^3 z^4$. Thus $x^3 z^2 \in \mathfrak{p}$ but $x, z \notin \mathfrak{p}$, contrary to primality.

If z^3 does not occur, then $z^3 f \in Q$, so primality requires $zf \in Q$. For a term $x^a y z^c$ with $a + c \leq 2$, its product by z belongs to Q only if $c \geq 2$, or if $a \geq 2$ and $c \geq 1$. The second alternative violates the degree bound; the first forces $a = 0, c = 2$. Hence f is a nonzero scalar multiple of yz^2 . But (2) gives

$$(Q : yz^2) = (y, z, x^2),$$

which contains x^2 and not x and therefore is not prime. This excludes every homogeneous polynomial of degree at most three and proves $v(Q) \geq 4$. Together with (3) and the closure formula, this completes the proof of Theorem 1.

4 Verification and scope

The accompanying Python verifier checks, in exact integer arithmetic, closure certificates at the 60 points of $[0, 3] \times [0, 2] \times [0, 4] \cap \mathbb{Z}^3$, the two upper colons, and all nonempty homogeneous supports outside the ideals in the degrees needed for the lower bounds: 39 for J in degrees 0–2 and 166 for Q in degrees 0–3. For each support it finds monomials u, v for which uf and vf have surviving terms, while every term of uvf belongs to the ideal. Such a certificate excludes a prime colon for every assignment of nonzero coefficients over every field. The unbounded closure formula is established by the proof in Section 2.

Zeng’s study of the whiskered-graph weak Lefschetz question [3] provides adjacent context for exact verification in monomial algebra. No theorem from that work is used here.

Computational and AI assistance. OpenAI Codex assisted with the counterexample analysis, exact verification scripts, proof checking, and manuscript preparation. A separate model context checked the mathematical argument and all 205 low-degree supports. This process did not involve external peer review or a proof assistant. Run `python3 manuscript/verify.py` from the accompanying source directory to replay the finite checks.

References

- [1] Prativa Biswas, Mousumi Mandal, and Partha Phukan, *A comparison of the v -number of a monomial ideal and its integral closure*, arXiv:2609.05044v1 (2026). <https://arxiv.org/abs/2609.05044v1>.
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